often contrast particles with waves, but what they really mean is particles and fields.

Quantum mechanics ultimately unified particles and fields into a single entity, the wave function. The impetus to do so came from two directions: first, physicists discovered that things they thought were waves, like the electric and magnetic fields, had particle-like properties. Then they realized that things they thought were particles, like electrons, manifested field-like properties. The reconciliation of these puzzles is that the world is fundamentally field-like (it's a quantum wave function), but when we look at it by performing a careful measurement, it looks particle-like. It took a while to get there.

Particles seem to be pretty straightforward things: objects located at particular points in space. The idea goes back to ancient Greece, where a small group of philosophers proposed that matter was made up of point-like "atoms," for the Greek word for "indivisible." In the words of Democritus, the original atomist, "Sweet is by convention, bitter by convention, hot by convention, cold by convention, color by convention; in truth there are only atoms and the void."

At the time there wasn't that much actual evidence in favor of the proposal, so it was largely abandoned until the beginning of the 1800s, when experimenters had begun to study chemical reactions in a quantitative way. A crucial role was played by tin oxide, a compound made of tin and oxygen, which was discovered to come in two different forms. The English scientist John Dalton noted that for a fixed amount of tin, the amount of oxygen in one form of tin oxide was exactly twice the amount in the other. We could explain this, Dalton argued in 1803, if both elements came in the form of discrete particles, for which he borrowed the word "atom" from the Greeks. All we have to do is to imagine that one form of tin oxide was made of single tin atoms combined with single oxygen atoms, while the other form consisted of single tin atoms combined with two oxygen atoms. Every kind of chemical element, Dalton suggested, was associated with a unique kind of atom, and the tendency of the atoms to combine in different